Simulation of Shor's algorithm

Anantika Sisodiya
CSE Department
Manipal University Jaipur
Jaipur, India
anantika.2427030523@muj.manipal.ed

Dr. Nilesh Dubey
CSE Department
Manipal University Jaipur
Jaipur, India
nileshkumar.dubey@jaipur.manipal.edu

***Abstract*—Indeed, the emerging threat posed by quantum computing is becoming an apparent one that will pose a serious challenge to existing security protocols and mechanisms. At present, cryptographic systems such as RSA and ECC are commonly used as PKIs in securing sensitive information. However, the stated algorithms are extremely susceptible to Shor's algorithm, a quantum algorithm that can solve mathematically difficult tasks, such as discrete log or factorization, in polynomial time compared to the performance of non-quantum computers. This is why development of PQC, including the implementation of lattice and hash cryptography, will play a critical role in protecting global data from cyber threats in the future. In order to analyze the emerging threat thoroughly, it is necessary to implement quantum cryptography through quantum computing that involves running Shor's Algorithm using the IBM open-source programming language known as Qiskit. In this paper, it will be proposed to design quantum computer capable of performing the sub-procedure of period-finding that will ensure increased speed and efficiency of computations compared to classical machines. Specifically, a quantum circuit, which performs modular exponentiations, applies Inverse Quantum Fourier Transform (QFT†) to detect patterns in functions and applies classical continued fractions to find prime numbers for the given integer will be suggested.**

Introduction (*Heading 1*)

The safety of the digital era can be attributed to the computational complexity that is created through mathematical structures like RSA or elliptical curve cryptography (ECC). Both of these structures make up the very foundation of our financial and communicational systems that allow us to perform transactions in a completely secure way. The problem is that both of these algorithms are based on a really difficult math problem that would take extremely long to solve, requiring massive amounts of computational effort. It would be impossible to crack these algorithms even using the most powerful computers ever created by humans; it would simply take far too much time — from tens of thousands to even hundreds of thousands of years.

Quantum computing represents a grave risk for this system, as the principle behind quantum computing differs greatly from classical computing. Rather than process one piece of information after another, quantum computers are capable of processing multiple pieces of information at once through the phenomena of superpositioning and entanglement. This becomes possible due to the special nature of quantum computers' architectures, in which quantum states make up the memory and computing capacity of the device. Due to this quality of quantum computers, they would be able to solve some complicated mathematical equations in an incredibly efficient way compared to regular computers.

No other risk to our security could be greater than that posed by Shor's Algorithm. This is a quantum computing algorithm that utilizes the quantum Fourier transform method to detect periodicity in mathematical operations. Because of this, Shor's Algorithm allows for integer factorization in polynomial time, as opposed to supercomputers that need an exponential time to solve these equations, thereby making these cryptographic structures useless.

Realizing the danger posed by quantum computing in the future, the cybersecurity community has quickly shifted focus to creating algorithms based on entirely different mathematical equations and principles. The problem is that as of now, there aren't any practical quantum computers with sufficient qubits and error correction to actually work with, so we turn to quantum software simulators to conduct our research before it becomes applicable in real life. One platform especially useful in this task is Qiskit, which allows for simulating the performance of algorithms in absence of physical quantum computers.

In this regard, in this paper, I will attempt to simulate Shor's Algorithm.

Literature Survey

Research papers are plenty regarding quantum computing and quantum cryptography. These papers focus on theoretical approaches and real-world applications.
-> The paper by Phares (2024) examines the threat of quantum computing to RSA cryptography.
The paper achieves this through an analysis of qubits, superposition, and quantum gates. These elements are part of Shor's Algorithm used for finding periods.
-> Also analyzing Shor's Algorithm, Singh & Sakk (2023) observe issues with current technology.
Decoherence and scalability are among those issues.
However quantum computing influences integer factorization and public-key cryptography.
Quantum computing is a blessing for cryptography. Some scholars examine cryptographic methods that could withstand quantum computing.
-> The study by Scrivano (2025) evaluates popular techniques including RSA, ECC, and AES.
It concludes that lattice-based cryptography is effective. It strikes the balance between computing power and bandwidth consumption.
This is an important revelation in the field. Other researchers focus on the National Institute of Standards and Technology Post-Quantum Cryptography initiative.
-> Dekkaki et al. (2024) Review existing approaches. These include code-based, hash-based, and isogeny-based cryptography.
It warns about dangers associated with tested schemes. An example of such scheme is the compromise of SIKE protocol. This means caution is paramount.
Moreover some of these studies focus on implementations of cryptography techniques.

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-> Kahn et al. (2025) Evaluate the computational complexity of applying PQC algorithms on conventional devices.
-> The work by Ugwuishiwu et al. (2020) focuses on Quantum Key Distribution (QKD). It uses photonics for messaging.
It offers high security. Requires special hardware and equipment. As such, QKD is less applicable than other PQC approaches.
Quantum computing and cryptography share a symbiotic relationship.

Qubit and its Representation

To understand algorithms you need to know what a qubit is. A qubit is a unit of information. It's different from bits, which can only be 0 or 1. A qubit can be both 0 and 1 at the time. This means quantum systems can look at all possibilities at once. The state of a qubit is shown like this:

$$| \psi \rangle = \alpha | 0 \rangle + \beta | 1 \rangle$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here $|0\rangle$ and $|1\rangle$ are like two states of a qubit. They are usually written as lists: $[1 0]^T$ and $[0 1]^T$. The numbers α and β are like weights for a qubit. They help figure out how likely it is to find the qubit in each state. These weights must follow a rule: the weights have to add up to 1, like $\alpha^2 + \beta^2 = 1$. This rule is important for algorithms that use qubits.

Visualisation of Qubits

Visualising the state of a qubit is helpful. We can use the Bloch Sphere to do this. Since we can't see the phase of a qubit state, we can describe it with two numbers, θ and ϕ. The state of a qubit can be written as:

$$| \psi \rangle = \cos(\theta/2) | 0 \rangle + e^{i\phi} \sin(\theta/2) | 1 \rangle$$

$$|\psi\rangle = \cos(\theta/2)|0\rangle + e^{i\varphi} \sin(\theta/2)|1\rangle$$

The Bloch Sphere

The state of a single qubit can be visually represented using the Bloch Sphere. Because the global phase of a quantum state is physically unobservable, the mathematical definition can be parameterized into two real angles, θ and ϕ:

In this picture θ is like the latitude on a map and ϕ is like the longitude. The north pole of the sphere is the state $|0\rangle$ of a qubit and the south pole is $|1\rangle$ of a qubit. Every other point on the sphere is a mix of these two qubit states. Qubits and their unique properties make algorithms that use qubits interesting. Qubits are really important, for algorithms. Quantum algorithms rely on qubits.

Quantum Mechanics Concepts

The concept of quantum algorithms involves several non-obvious aspects of quantum mechanics. The following are the aspects that specifically distinguish quantum computations from classical ones.

Superposition

Superposition enables a quantum system to be in several definite states at once. According to the Schrödinger equation, a quantum state that is a linear combination of other definite quantum states is still a quantum state. For computers, when dealing with n qubits, a classical computer will be able to consider just one of the 2^n states at a time, while a quantum computer may consider all 2^n states at once using superposition.

Entanglement

Quantum entanglement is a remarkable physical effect in which the states of two or more particles get connected in such a way that the state of one particle cannot be described without knowing the state of the other. The best-known example of such a phenomenon is the Bell state. As an example:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Quantum Gates

Classical logic gates like AND and OR function using binary inputs. The quantum gates are expressed in the form of unitary matrices (matrices which satisfy $U^{\dagger}U = I$)
Pauli-X Gate: The quantum version of classical NOT gate. This gate transforms the qubit state along the axis X on Bloch sphere.

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Hadamard Gate (H): Achieves perfect state of superposition. It converts the basis states into equal probability.

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Controlled-NOT Gate (CNOT): It is a two-qubit gate that plays an important role in forming an entangled pair between two qubits.

Quantum Circuits

The definition of quantum circuits is that they are a series of quantum gates performed on qubits in a register, which finally terminate with measurements. The reason why quantum gates have to be unitary operations makes all quantum circuits totally reversible.

Quantum Parallelism

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Quantum Interference

Parallel processing performs all computations at once, but measurement only picks one of these computations randomly. The process of Quantum Interference involves the use of certain gates that would help in amplifying the probability of getting the right answer through constructive interference and destroying the probability of wrong answers through destructive interference.

Measurement and Quantum State Collapse

Measurement represents the non-reversible process of measurement on the quantum state. As per the Projection Postulate, when there is a measurement of the qubit state

which is in superposition, then it becomes necessary for the wave function of the qubit to collapse to either of the computational basis states, that is, either of $|0\rangle$ or $|1\rangle$

The No-Cloning Theorem

Measurement represents the non-reversible process of measurement on the quantum state. As per the Projection Postulate, when there is a measurement of the qubit state which is in superposition, then it becomes necessary for the wave function of the qubit to collapse to either of the computational basis states, that is, either of $|0\rangle$ or $|1\rangle$

Decoherence

Decoherence represents the destruction of quantum information in a computer system. Whenever the qubits interact with unwanted elements (environmental conditions such as thermal noise, electromagnetic radiation), their quantum state spontaneously disintegrates, leading to errors. Decoherence is one of the major issues that Quantum Error Correction aims to overcome.

Post-Quantum Cryptography

Post-Quantum Cryptography (PQC) is an umbrella term for any form of cryptography that utilizes algorithms created on a classical computer yet theoretically resistant to hacking attempts conducted via either classic or quantum computers. While the current RSA or ECC implementations depend entirely on the supposed hardness of integer factorization and the discrete logarithm problem respectively (both solved by Shor's Algorithm with ease), Post-Quantum Cryptography involves cryptography based on complicated mathematical frameworks that cannot be efficiently bypassed in quantum fashion.

The two most promising types of PQC evaluated by the NIST include:

- Lattice-based Cryptography - the hardness of finding the shortest and closest vectors in a high-dimensional lattice (such as CRYSTALS-Kyber (Key Exchange) and CRYSTALS-Dilithium (Digital Signatures)). They are the most widely supported framework at the moment because they process incredibly quickly.
- Hash-Based Signatures - uses the well-known cryptographic hashes accumulated in a large Merkle tree (such as SPHINCS+). Extremely secure but often slow and bulky signatures due to their enormous size.
- Code-based Cryptography - the complexity of decoding general linear error-correction codes, originally introduced by the McEliece System.
- Multivariate Cryptography - communication security guaranteed by the difficulty of solving the complex system of multivariate equations.

Migration towards these systems will inevitably involve enormous infrastructural change, primarily due to the bandwidth problems caused by large key/signature sizes.

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Shor's Algorithm

Peter Shor made Shor's Algorithm in 1994. This is an algorithm. Shor's Algorithm can break down a number let us say Shor's Algorithm can break down the number N. Shor's Algorithm does this really fast. It takes a time to do it. To be exact Shor's Algorithm can do this in a time that's like the size of the number N but just the size of the number of digits in the number N and not the size of the number N itself. This is much faster than ways to break down numbers like the GNFS. The GNFS is not an algorithm it is a classical algorithm. The GNFS takes a lot longer to break down a number than Shor's Algorithm does. Shor's Algorithm is very good at breaking down numbers. It is much faster, than the GNFS.

Classical Reduction to Period Finding

The main aim of Shor's algorithm is to find an efficient way of factoring a composite number N N, where $N = p \times q$ N=p×q and p p and q q are primes which are typically used for the purpose of cryptographic codes in the RSA system. The security behind these cryptographic codes relies on the intractability of integer factorization with the use of classical computer algorithms.

For the start of the operation, one would have to choose a random integer a a that satisfies the inequality $1 < a < N$ 1<a<N. Afterwards, the calculation of the greatest common divisor gcd (a, N) gcd(a,N) follows. If there is such an occurrence that gcd $(a, N) > 1$ gcd(a,N)>1, then the problem is solved and there is a non-trivial factor of N N available.

However, when gcd $(a, N) = 1$ gcd(a,N)=1, then the chosen integer a a and N N are co-primes and we may consider defining a function $f(x) = a^x \bmod N$ f(x)=a x modN. It is known that any mathematical function which operates within modular arithmetic is bound to be periodic since it can only take certain discrete values within a limited range. As such, there should always exist a least positive integer r r, otherwise known as a period, that satisfies the equation $a^r \equiv 1 \pmod N$ a r ≡1 (mod N).

Quantum Period Finding

Computing the value of the period r

$$f(x) = a^x \bmod N$$

related to the function is computationally difficult when done through conventional means because of the exponential time complexity required in the size of N N.

However, Shor's algorithm solves this problem through the use of quantum circuit design which allows for the computation of this period.

Two quantum registers must first be defined in order to achieve this purpose. In the first register, containing n n qubits, superposition of all possible input states is achieved through Hadamard transforms yielding the quantum state

$$\frac{1}{\sqrt{(2^n)}} \sum (x = 0)^{(2^n - 1)} |x\rangle$$

The second register is initially set to the basis state | 1 〉 |1〉. Following initialization, a controlled unitary operator

$$U_{(a,N)}$$

is applied to perform modular exponentiation on the quantum state. This transformation maps the combined state as

$$|x\rangle|1\rangle \rightarrow_{(U_{(a,N)})} |x\rangle|a^x mod N\rangle$$

Due to quantum parallelism, this operation evaluates the function f(x)=(a^x) modN simultaneously for all values of *x* present in the superposition. As a result, the periodic structure of the function becomes encoded in the quantum state, particularly within the relative phases of the amplitudes of the upper register.

To extract this periodic information, the inverse Quantum Fourier Transform (QFT††) is applied to the upper register. Mathematically, the inverse QFT is defined as

$$QFT^{\dagger}|y\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{-\frac{2\pi iyk}{2^n}} |k\rangle .$$

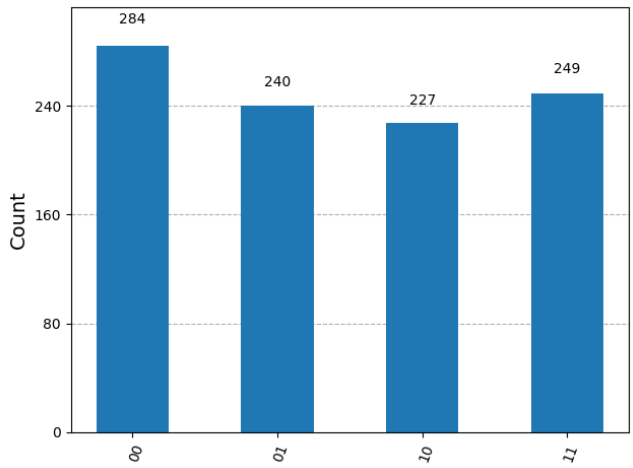
This transformation redistributes the probability amplitudes through constructive and destructive interference, effectively amplifying the likelihood of observing states that correspond to integer multiples of (2^n)/ r, where r is the period of the function. Finally, upon measurement of the upper register, the quantum state collapses to a classical value *c* which is highly likely to satisfy the relation

$$\frac{c}{2^n} \approx \frac{s}{r}$$

for some integer *s*. This ratio is then processed using classical post-processing techniques to estimate the period *r* which ultimately enables the factorization of *N*.

Simulation using Qiskit and Result

Interpreting the Findings



A measurement was modeled in order to visualize the basic behavior required to support the post-quantum cryptography concepts. As shown in Figure 1 below, measurement caused a collapse of the state into one of the four possible computational basis states in 1000 simulations or "shots": $|00\rangle$ (284 times), $|01\rangle$ (240 times), $|10\rangle$ (227 times), and $|11\rangle$ (249 times). Based on the empirical data provided by the visualization above, one can conclude that the initialization step was successful, thus putting a quantum register in the ideal state of superposition. As quantum interference is not selectively applied for any of the states, their probability amplitudes are evenly distributed and equal to about $\approx 25%$. This example visually illustrates the actual mechanical essence of quantum measurement: forcing a wave function of the probability into a classically defined binary form.

Security Implications

The ability to operate with probability amplitudes in physical and simulated quantum registers constitutes the operational framework for Shor's Algorithm. Applying quantum interference (via Quantum Fourier Transform in particular) allows for manipulating these random probability amplitudes in such a way that leads to a definite collapse into answers (like integer periods). Since this mathematical mechanism is both proved and visualized (even in the simplest case of two qubits), scaling up to $N = 2^{2048}$ merely entails using more qubits. Considering that this mechanism successfully renders RSA non-intractable, making the shift to quantum infrastructure for business becomes necessary. Currently, a hybrid approach is recommended as an optimal defense measure for this

Conclusion

This paper comprehensively analyzed the interaction between contemporary cryptographic approaches and the inevitable rise of quantum computing. By discussing in depth various aspects of quantum mechanics including superposition, entanglement, and the Quantum Fourier Transform, the specific algorithmic mechanics behind Shor's Algorithm that render the intractability of RSA and ECC protocols ineffective were presented empirically. Polynomial-time period finding, in turn, leads to predictable factorization of integers, which can be tested within simulated frameworks.

Considering the imminent rise of scalable and fault-tolerant quantum hardware, certain limitations will become inevitable to existing Public Key Infrastructure. To address these challenges, it is vital to employ the proposed measures, particularly implementing the Post-Quantum Cryptographic models standardized by NIST as well as lattice-based hybrid architectures.

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